AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T216B
Cisco Networking (CCCNA) part B

Midterm Examination (MTA)
First Semester 2025/2026

Date: 20/Nov./2025

Number of Exam Pages: 6 (including this cover sheet)	**Time Allowed:** 2h

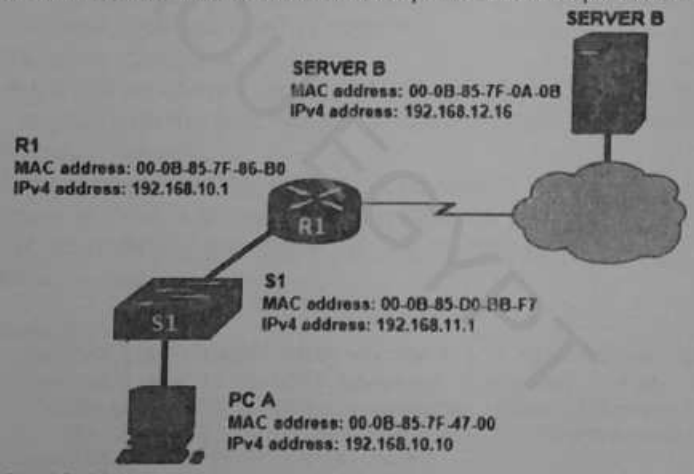
Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part 1: Multiple choices questions- 1 mark per question **Total: 5 Marks**

1. A technician is configuring the channel on a wireless router to either 1, 6 or 11. What is the purpose of adjusting wireless channel in WLAN?
   a. To enable different 802.11 standards
   b. To provide stronger security modes
   c. To avoid interference from nearby wireless devices
   d. To disable broadcasting of the SSID
2. What device is considered as supplicant during the 802.1X authentication process?
   a. The router that is serving as the default gateway
   b. The client that is requesting authentication
   c. The authentication server that is performing client authentication.
   d. The switch that is controlling network access
3. A network administrator is analyzing the features that are supported by different first-hop router redundancy protocols. Which statement describes a feature that is associated with HSRP?
   a. It uses ICMP messages in order to assign the default gateway to hosts.
   b. HSRP uses active and standby routers.
   c. It allows load balancing between a group of redundant routers.
   d. HSRP is nonproprietary.
4. Refer to the beneath image. PC A sends a request to Server B. What IPV4 address is used in the destination field in the packet as the packet leaves PC A?



   a. 192.168.10.1
   b. 192.168.12.16
   c. 192.168.10.10
   d. 192.168.11.1
5. An administrator issues the **ipv6 route 2001:acad:1::/32 gigaethernet 0/0 2001:db8:acad:6::1 100** on a router. What administrative distance is assigned to this route?
   a. 0
   b. 100
   c. 32
   d. 1

Part 2: Answer all the questions – 4 marks per question **Total: 16 Marks**

1. Answer the following questions:
   a. Explain what will happen when the forwarding router, in the First Hop Redundancy Protocols (FHRPs), fails. (2 marks)

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b. Refer to figure B. Which destination MAC address is used when frames are sent from the workstation to the default gateway? (2 marks)

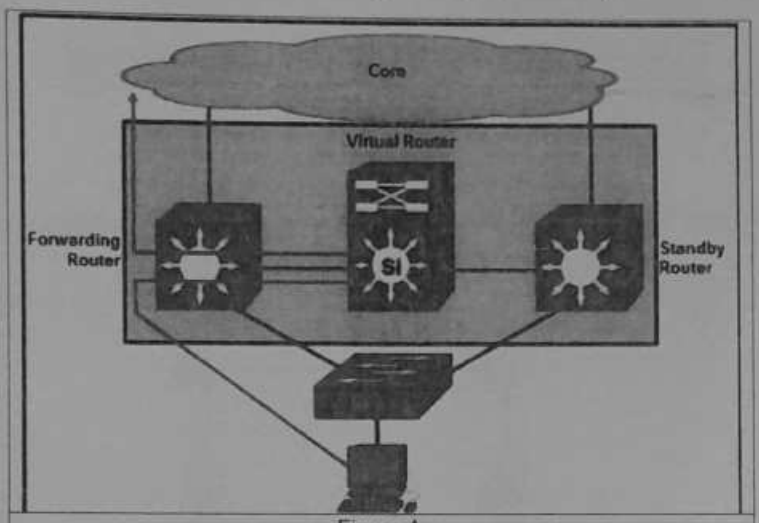


Figure A

2. Answer the following questions:
   a. When security is a concern, Which OSI layer is considered to be the weakest link in a network system? (2 marks)
   b. A network administrator of a college is configuring a AAA server for his WLAN. What it the reason behind this? (2 marks)
3. Answer the following questions:
   a. What is the purpose of the Dynamic ARP Inspection (DAI)? (2 marks)
   b. How can we mitigate the MAC Address Table attack? (2 marks)
4. Answer the following questions:
   a. What is the purpose of the floating static router? (2 marks)
   b. Refer to figure B. What is the administrative distance value of the route for router R2 to reach the destination IP address of 192.168.10.200? (2 marks)

```
R2#show ip route

<output omitted>
D    192.168.1.0/24 [90/2172416] via 200.0.0.1, 00:02:00, Serial0/0/0
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/1
L       192.168.2.1/32 is directly connected, GigabitEthernet0/1
S    192.168.10.0/24 [1/0] via 197.10.0.1
D    192.168.20.0/24 [90/2297856] via 200.0.0.1, 00:02:00, Serial0/0/0
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, Loopback3
L       192.168.30.1/32 is directly connected, Loopback3
     192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.40.0/24 is directly connected, Loopback4
L       192.168.40.1/32 is directly connected, Loopback4
     197.10.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       197.10.0.0/24 is directly connected, GigabitEthernet0/0
L       197.10.0.2/32 is directly connected, GigabitEthernet0/0
     200.0.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.0.0.0/30 is directly connected, Serial0/0/0
L       200.0.0.2/32 is directly connected, Serial0/0/0
```

Figure B

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Part 3: Answer all the following questions **Total: 29 Marks**

1. Answer the following questions (10 marks)

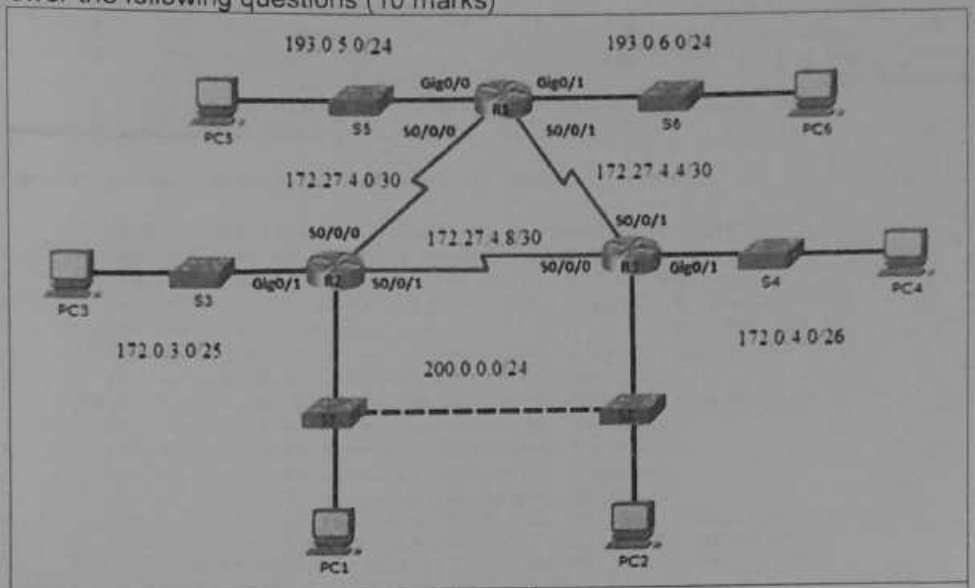


Figure 1

a) How many networks are directly connected to R3? (1 mark)
b) How many remote networks are there for R3? (1 mark)
c) On router R3, write the command to configure a: (4 marks)
   - static route to 172.0.3.0/25 using the next hop IP address 172.27.4.9
   - static route to 193.0.5.0/24 using the exit interface S0/0/1
d) On Router R3, write the command to configure a floating static route to 172.0.3.0/26 using the exit interface S0/0/0. (2 marks)
   Note: the administrative distance is 11
e) On router R2, write the command to configure a default static route to using the next hop P address 172.27.4.1(2 marks)

2. Figure 2 displays the routing table of router R1. (9 marks)
   Refer to figure 2 and answer the following questions:
   a) How many directly connected networks does R1 have? (1 marks)
   b) List all the directly connected networks for R1. (2 marks)
   c) How many networks were learned dynamically by R1? (1 marks)
   d) List all the networks that were learned dynamically by R1. (3 marks)
   e) Via which interface will R1 forward the packet if the destination address is:
      e1)200.0.1.150
      e2)190.30.1.99
      (2 marks)

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```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 120.0.0.2 to network 0.0.0.0

     120.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       120.0.0.0/30 is directly connected, Serial0/0/0
L       120.0.0.1/32 is directly connected, Serial0/0/0
O       120.0.0.4/30 [110/845] via 120.0.0.2, 00:10:53, Serial0/0/0
O       120.0.0.8/30 [110/845] via 120.0.0.13, 00:10:53, Serial0/0/1
C       120.0.0.12/30 is directly connected, Serial0/0/1
L       120.0.0.14/32 is directly connected, Serial0/0/1
     190.30.0.0/16 is variably subnetted, 4 subnets, 2 masks
C       190.30.1.0/24 is directly connected, Loopback1
L       190.30.1.1/32 is directly connected, Loopback1
C       190.30.2.0/24 is directly connected, Loopback2
L       190.30.2.1/32 is directly connected, Loopback2
O       200.0.1.0/24 [110/65] via 120.0.0.2, 00:10:53, Serial0/0/0
O       200.0.2.0/24 [110/65] via 120.0.0.2, 00:10:53, Serial0/0/0
O    220.220.0.0/24 [110/65] via 120.0.0.13, 00:10:53, Serial0/0/1
O*E2 0.0.0.0/0 [110/1] via 120.0.0.2, 00:10:53, Serial0/0/0

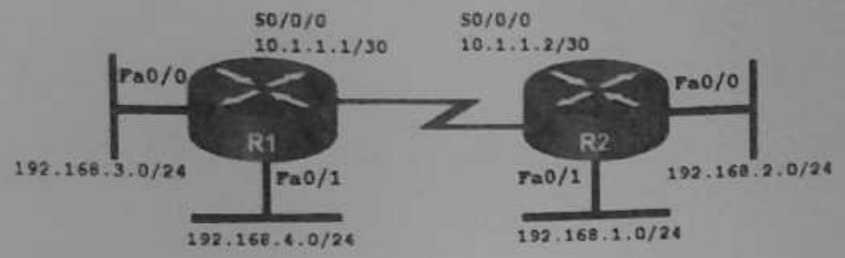
R1#
```

Figure 2

3. Answer the following questions: (10 marks)
   a) Write the commands needed to configure port security on interface G0/2 of switch S1. (4 marks)
      To do this complete the following tasks:
      - Enable port security on interface G0/2 (2 marks)
      - Configure the switch to allow only two mac addresses on this interface
      - Configure the switch to dynamically learn the MAC addresses and "stick" them to the running configuration (1 mark)
      - Set the security violation mode to restrict (1 mark)

      Note: you are already in the interface sub-configuration mode of G0/2
   b) Refer to figure 3. A ping from R1 to 10.1.1.2 is successful, but a ping from R1 to any address in the 192.168.2.0 network fails. What is the cause of this problem? (3 marks)

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```
R1# show ip route

<output ommited>
Gateway of last resort is not set
10.0.0.0/30 is subnetted, 1 subnets
C   10.1.1.0 is directly connected, Serial0/0/0
D   192.168.1.0 [120/1] via 10.1.1.2 00:03:50, Serial0/0/0
C   192.168.3.0/24 is directly connected, FastEthernet0/0
C   192.168.4.0/24 is directly connected, FastEthernet0/1
S   192.168.2.0/24 [1/0] via 192.168.3.0
```

Figure 3

c) The network administrator has configured HSRP on both router R1 and R3 (figure 4). Router R1 was configured as the standby router and R3 as the active router. However, the administrator noted that R1 is always the active router even after rebooting the two routers. Refer back to the R1 and R3 configuration of figure 4 and explain why R1 is always the active router. (3 marks)

Router R1	Router R3
R1#show running-config Building configuration... Current configuration : 1187 bytes ! output omitted! ! hostname R1 ! output omitted! ! interface GigabitEthernet0/0 ip address 172.159.0.2 255.255.255.0 duplex auto speed auto standby version 2 standby 9 ip 172.159.0.19 standby 9 preempt !	R3#show running-config Building configuration... Current configuration : 1094 bytes ! output omitted! ! hostname R3 ! output omitted! ! interface GigabitEthernet0/1 ip address 172.159.0.3 255.255.255.0 duplex auto speed auto standby version 2 standby 9 ip 172.159.0.19 standby 9 priority 90 standby 9 preempt !

Figure 4

End of questions

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